Digital Phenotyping: Development and Validation of a Robust Colour Correction Pipeline

Graduation Project (PFE)

Vilmorin-Mikado (Limagrain Group)

National School of Computer Science (ENSI)

Presented by	Academic Supervisor
Aymen Masmoudi	Dr. Majdi JRIBI

Professional Supervisor

Mr. Cyril DAMBRINE



Aymen Masmoudi

Final-year IT Engineering student at ENSI, specializing in **Computer Vision and Data Science**. Passionate about applying image processing and machine learning to solve real-world challenges.

Degree

IT Engineering — ENSI

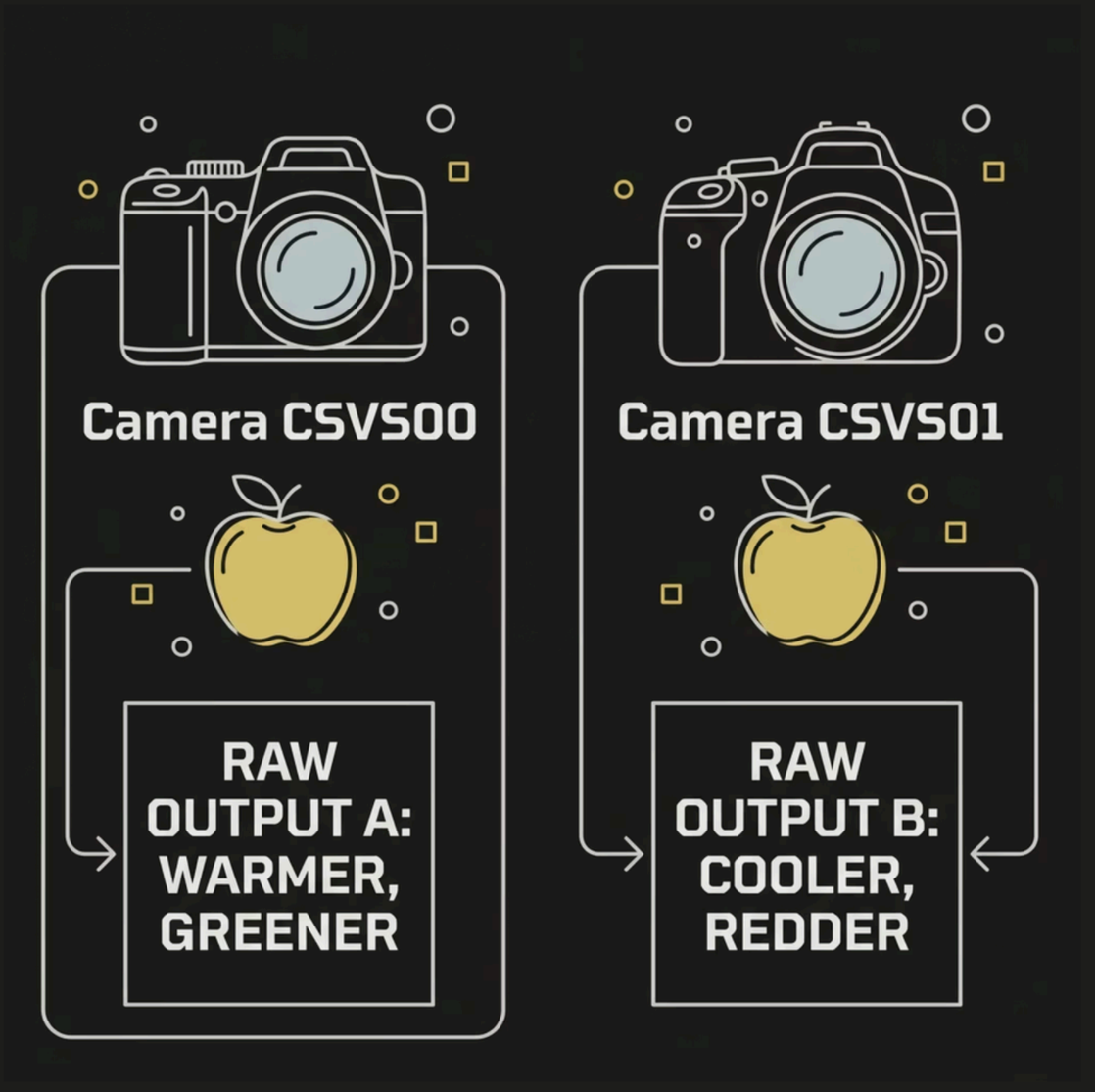
Specialization

Computer Vision & Data Science

Project Context & Problem Statement

Phenotyping relies on **precise color measurement** to assess fruit ripeness and quality. However, different CSVS imaging devices have different sensor responses, causing identical objects to appear differently in raw data across cameras.

A standardized, device-independent color pipeline is essential for reliable, reproducible phenotyping data.



Project Objectives

The pipeline targets a **perceptually imperceptible** color deviation between devices, quantified by the international standard *Delta E_2000 < 2*.

1

Primary Goal

Implement a modular image-processing pipeline that performs segmentation, ROI extraction, pixel clustering, color correction

2

Color Correction Pipeline

Reduce inter-device color deviation to Delta E < 2 the threshold of human perceptual imperceptibility

3

Temporal Stability

Analyze color drift over time and establish a standardized correction protocol.

4

Cross-fruit Validation

Validate the pipeline across multiple fruit types: **Pepper** and **Zucchini**.

Project Roadmap

A structured five-phase timeline spanning from initial familiarization through final documentation and presentation.

March 1

Introduction, familiarization, and experimentations

2 **April**

Temporal stability analysis and large-scale multi species dataset processing.

May 3

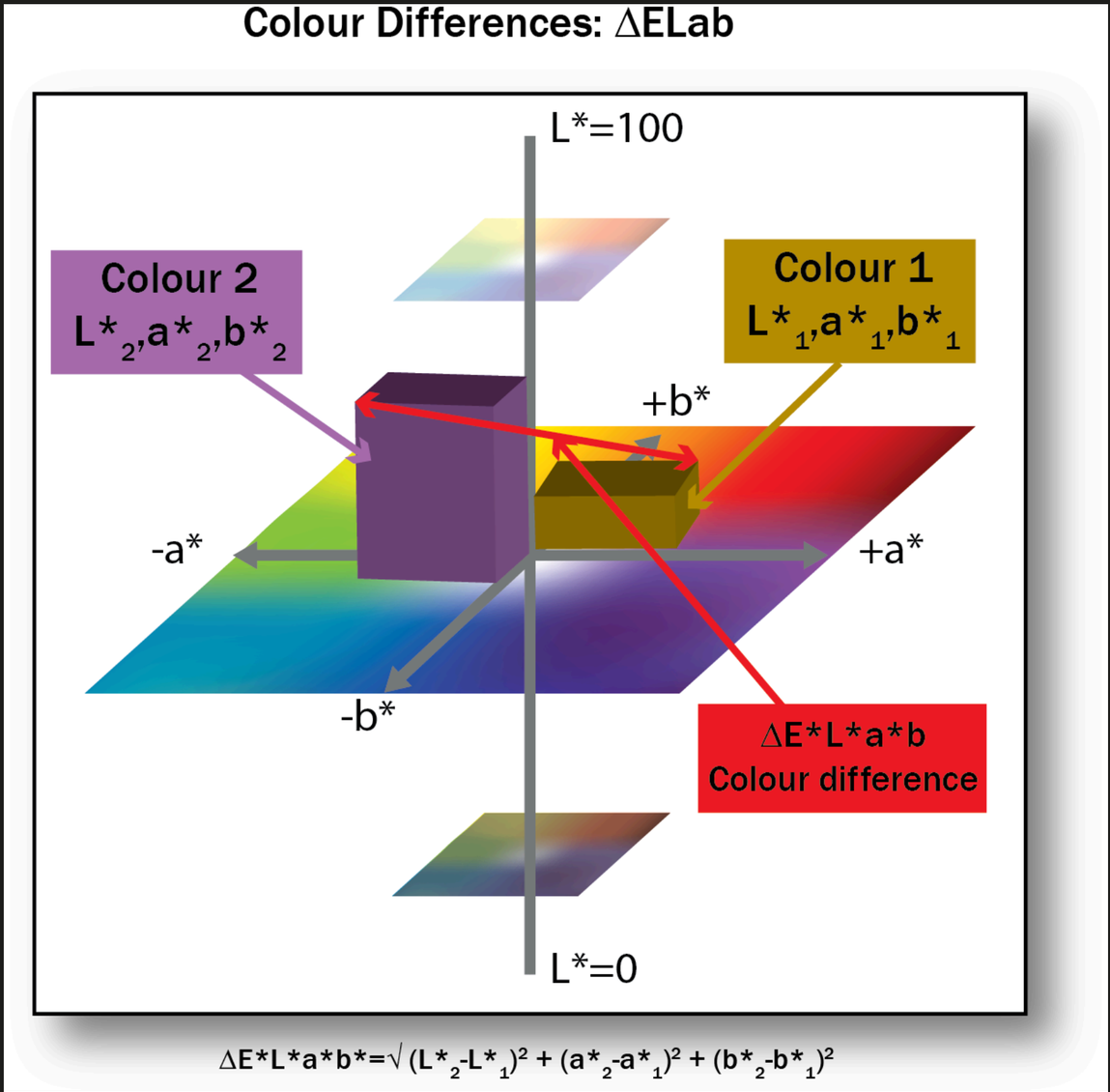
Pipeline development, preliminary testing, and color correction validation.

4 **June**

Analysis , Validation , refinement and multi-device cross-validation.

July – August 5

Final documentation, report writing, and presentation preparation.



Key Terminology

CIELAB (*L^*a^*b^**)

Color space approximating human vision. **L** = Lightness, **a** = Green–Red axis, **b** = Blue–Yellow axis.

Delta E_2000

Standard perceptual metric quantifying the difference between two colors. Values below 2 are imperceptible to the human eye.

CC / NOCC

CC: Color Correction via transformation matrix or RPCC.
NOCC: Raw, uncorrected image data.

ROI

The fruit surface area extracted from the image for color analysis.

Global Methodology: The Pipeline



A six-stage end-to-end workflow processes raw camera files into validated, device-independent color measurements implemented as modular Python notebooks, enabling reproducible batch processing across multiple devices and fruit datasets.

Dataset Restructuring (Pepper)

The pepper dataset is consisted of **209 matched image pairs** captured across two devices. Images from multiple sessions were organized into a unified directory structure for automated batch processing.

Implemented a device deviation comparison notebook
Made statistical analysis on cc and device deviation



Red
76 pairs

Yellow
89 pairs

Orange
44 pairs

Validation Results: (Pepper)

Colour	mean NOCC μ	mean CC μ	Median NOCC μ	Median CC μ	Shift
ORANGE	3.928	2.255	4.688	2.791	Large left shift clear improvement
RED	2.474	1.484	3.020	2.074	Large left shift best result
YELLOW	3.536	3.012	4.073	3.489	Moderate shift insufficient

Before Correction (NOCC)

Global Mean *Delta E = 3.23* clearly perceptible inter-device difference.

After Correction (CC)

Global Mean *Delta E = 2.29* a **29% improvement** overall.

Red: *Delta E = 1.48*

Excellent , below the imperceptibility threshold of 2.0.

Yellow: *Delta E = 3.01*

Further cc refinement required for this color class.

Current Task: Temporal Stability

Investigating sensor performance over time to ensure consistent color measurements for robust phenotyping.

Objective

is there an intra or inter drift during different times in the day ?

Experiment Design

Three repetitions captured at 30-minute intervals.

Analysis Points

Inter-Device: CSVS00 vs CSVS01 at each time point.

Intra-Device: CSVS00 at rep 1 vs CSVS00 at rep 2.

Initial Findings

no results yet

ROI Extraction & CC for Zucchini

1

Goal

Adapting the color correction pipeline developed for peppers to the zucchini dataset.

2

Innovation

Implementing automated window detection for Lucideon color patches, streamlining the calibration process.

3

Metrics

Tracking *Delta E* and Euclidean distance in 16-bit space to quantify color deviation and accuracy.

4

Status

to be completed



Software Tools & Environment

Our robust color correction pipeline is built upon a foundation of cutting-edge open-source software and libraries, ensuring accuracy and reproducibility.



Python 3.x

The primary programming language for developing and orchestrating the entire pipeline, offering flexibility and extensive scientific computing capabilities.



Colour-science

Essential for precise CIE color space transformations and the accurate calculation of Color Transformation Matrices (CTM) for correction.



Rawpy

Enables professional-grade decoding and linearization of raw camera image files (.ARW, .DNG), providing high-fidelity data input for the pipeline.



Pandas & Matplotlib

Utilized for comprehensive statistical validation of experimental results and powerful, clear data visualization capabilities to interpret findings.



OpenCV

Leveraged for advanced image processing tasks, including automated Region of Interest (ROI) and color patch detection, critical for calibration.

Summary of Accomplishments

Our rigorous approach yielded significant progress in developing a robust and reliable color correction pipeline for phenotyping applications.

Pipeline Developed

Successfully designed and implemented a complete, automated color correction pipeline, transforming raw image data into device-independent color measurements.

Quantified Deviations

Precise quantification of inter-device color deviations between CSV500 and CSV501 sensors (Task 2), establishing a baseline for improvement.

Error Reduction

Achieved a substantial 29% reduction in overall color error (ΔE) across the initial pepper dataset after applying the correction.

Sensor Limitations Identified

Pinpointed specific sensor limitations, particularly a persistent *L^** axis offset, impacting yellow samples and guiding future calibration efforts.

Preliminary Testing Completed

Successfully conducted preliminary analyses on temporal stability and lighting homogeneity, laying the groundwork for further optimization.

Next Steps & Recommendations

Building on our current findings, the immediate next steps involve refining our calibration methods and preparing the pipeline for broader application.



Target Upgrade

Implement a calibration target with a wider range of gray patches to improve the *L^** axis calibration, particularly for yellow hues.



Hardware Adjustment

Test the color correction pipeline under a controlled Lighting Dome environment to ensure perfectly uniform light distribution and eliminate bias.



Temporal stability

Continue investigating sensor performance over time to ensure consistent color measurements for robust phenotyping.



Final Validation

Complete the comprehensive analysis for the full Zucchini dataset, applying the refined pipeline to validate its performance across different plant types.

Conclusion & Questions

> "By standardizing the roi & color correction pipeline, we ensure that digital phenotyping remains a reliable, scientific tool for seed development, regardless of which camera or time of day is used."

Questions?